

Problem Statement

Evaluate the following limit from the file “dailyintegral-limit-limit-1x1-2026-07-18.jpg”:

$$\lim_{n \rightarrow \infty} \int_{-1}^1 \frac{x^{2n} + n}{1 + x^2 n^2} dx$$

Detailed Solution

This limit can be evaluated by exploiting the symmetry of the integrand and splitting the integral into two manageable parts. Let's break this down step-by-step.

Step 1: Exploit symmetry

Let $f(x) = \frac{x^{2n} + n}{1 + x^2 n^2}$. Notice that replacing x with $-x$ leaves the function unchanged, meaning $f(x) = f(-x)$. Since the integrand is an even function and the limits of integration are symmetric about zero, we can rewrite the integral as twice the integral from 0 to 1:

$$I_n = \int_{-1}^1 \frac{x^{2n} + n}{1 + x^2 n^2} dx = 2 \int_0^1 \frac{x^{2n} + n}{1 + x^2 n^2} dx$$

Step 2: Split the integral

By separating the terms in the numerator, we can split this into two distinct integrals:

$$I_n = 2 \int_0^1 \frac{x^{2n}}{1 + x^2 n^2} dx + 2 \int_0^1 \frac{n}{1 + x^2 n^2} dx$$

Step 3: Evaluate the first integral

Let's analyze the first term: $2 \int_0^1 \frac{x^{2n}}{1 + x^2 n^2} dx$. For all $x \in [0, 1]$ and $n \geq 1$, the denominator satisfies $1 + x^2 n^2 \geq 1$. Therefore, we can bound the integrand:

$$0 \leq \frac{x^{2n}}{1 + x^2 n^2} \leq x^{2n}$$

Integrating this inequality from 0 to 1 gives:

$$0 \leq \int_0^1 \frac{x^{2n}}{1 + x^2 n^2} dx \leq \int_0^1 x^{2n} dx$$

Evaluating the upper bound, we get:

$$\int_0^1 x^{2n} dx = \left[\frac{x^{2n+1}}{2n+1} \right]_0^1 = \frac{1}{2n+1}$$

Taking the limit as $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0$. By the Squeeze Theorem, the first integral evaluates to 0:

$$\lim_{n \rightarrow \infty} 2 \int_0^1 \frac{x^{2n}}{1+x^2n^2} dx = 0$$

Step 4: Evaluate the second integral

Now, consider the second term: $2 \int_0^1 \frac{n}{1+x^2n^2} dx$. We can evaluate this using the substitution $u = nx$. This means $du = n dx$, or $dx = \frac{du}{n}$. The limits of integration change from $x \in [0, 1]$ to $u \in [0, n]$. Substituting these yields:

$$2 \int_0^n \frac{n}{1+u^2} \frac{du}{n} = 2 \int_0^n \frac{1}{1+u^2} du$$

This is a standard integral that evaluates to the arctangent function:

$$2 [\arctan(u)]_0^n = 2(\arctan(n) - \arctan(0)) = 2 \arctan(n)$$

Step 5: Compute the final limit

Now we take the limit of this evaluated second term as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} 2 \arctan(n) = 2 \left(\frac{\pi}{2} \right) = \pi$$

Combining the limits from both parts, we arrive at our final result:

$$L = \lim_{n \rightarrow \infty} I_n = 0 + \pi = \pi$$

Final Answer:

$$\pi$$